AirGuard India: Station Calibration and Uncertainty-Aware Day-Ahead AQI Forecasting

Nishanth R

***Abstract*—Day-ahead air-quality forecasts require station-scale verification and an explicit distinction between observations and modelled concentrations. AirGuard India combines quality-controlled CPCB-provider measurements, archived operational CAMS forecasts and a fixed 20:00 IST issue-time contract to study station calibration, predictive uncertainty and exposure advisories. The design compares climatology, persistence and raw CAMS with direct and residual Ridge, LightGBM and XGBoost models using chronological and spatial validation. Severe AQI is retained as an open-ended lower bound rather than an exact regression label. A separate non-CAMS preliminary study finds a paired MAE difference of -8.96 AQI units for strict Ridge versus climatology (95% date-block interval: -12.49 to -5.61), while history Ridge does not establish improvement over persistence. In its first fold, both 80% and 90% intervals fail to cover any of 35 exact Very Poor outcomes. These findings motivate severity-specific calibration checks and transparent under-warning analysis. The genuine CAMS archive is partially acquired; primary correction results remain pending and the final test remains sealed.**

***Index Terms*—air quality forecasting, CAMS, station calibration, conformal prediction, exposure advisories**

I. Introduction

A useful next-day AQI forecast must describe what could have been issued before the target day and how well that forecast agrees with measurements. This distinction matters when an atmospheric model represents a grid cell, while a resident or monitoring instrument represents a particular location. Indian high-resolution pollution forecasting already has an established research context [1]. The present question is whether India-specific station correction improves a genuine operational CAMS baseline, and whether its uncertainty can support defensible warning choices.

AirGuard India fixes issuance at 20:00 IST for the next IST calendar day. Its retrospective stream reconstructs measured targets and admissible forecast features; its prospective stream freezes forecasts when actually issued. A past model field without its producing run cannot establish an operational forecast, and a deployed dashboard cannot establish predictive skill. Both distinctions shape the evaluation.

The work provides three research components:

- a versioned station dataset with recorded rejection reasons, causal history features and a shared CPCB-style AQI engine;
- a locked comparison of raw CAMS, direct learning and residual correction with rolling and spatial validation and censored-Severe diagnostics;
- an uncertainty and advisory evaluation that reports coverage failures, under-warning and over-warning, alongside immutable prospective verification.

The second and third components are evaluation commitments rather than completed primary findings. This manuscript distinguishes existing measured-data evidence from pending CAMS experiments and makes no world's-first claim.

Nishanth R is with the Department of AI and Data Science, Rajalakshmi Engineering College, Chennai, Tamil Nadu, India. Email: nishanth.r.2024.aids@rajalakshmi.edu.in; backup: nishanth.artwork@gmail.com.

Development draft. Primary CAMS results are pending; the final test remains sealed.

II. Related Work

A. Atmospheric Forecast Post-processing

Station-specific model output statistics is established. Riccio and Chianese study statistical improvement of CAMS with spatial verification [2]; Petetin et al. examine the trade-off between continuous and categorical ozone-forecast skill [3]. Operational regional CAMS production also includes statistical post-processing [4]. Consequently, introducing a correction learner alone does not establish novelty. AirGuard's contribution under investigation is the audited Indian station evaluation and its warning behaviour.

B. Uncertainty and Warning Decisions

Conformal methods provide a framework for data-driven prediction intervals. Time-series and distribution-shift settings require attention to dependence and changing coverage [5], [6]. A narrow interval or near-nominal aggregate coverage is insufficient if high-pollution categories are systematically missed. Proper forecast evaluation also separates the quantity predicted from the decision made from it [7]. Here, vulnerability groups specify exposure-warning tolerances; they do not constitute clinical cohorts or individual health-risk models.

C. Scope of the Comparison

Published methods use different targets, locations and verification periods. Their published scores are therefore not inserted into AirGuard's numerical leaderboard. The shared-observation CAMS comparison, spatial inference regimes and uncertainty failures must be established with this study's own artifacts. A complete novelty assessment remains pending the primary study.

III. Proposed Method

A. Issue-time Data Contract

For station s and target day d, $X_{s,d}$ is assembled before issue time, whereas $y_{s,d}$ is constructed later from measured

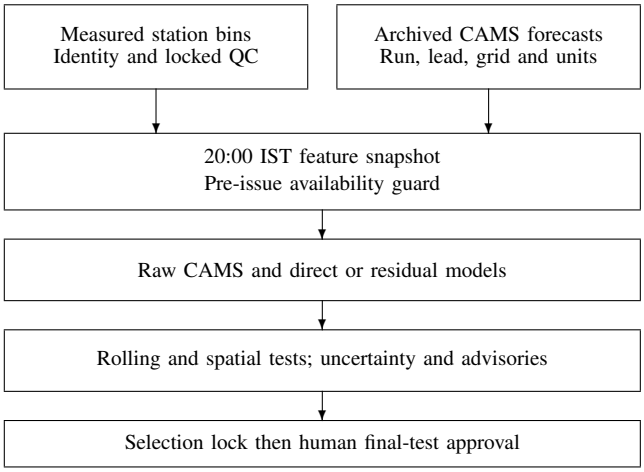


Fig. 1. AirGuard study architecture. The primary model comparisons and selection lock remain pending; prospective forecasts are a separate immutable stream. Targets are not used to choose issue-time predictors.

pollutant concentrations. Observations may contribute only when their bins end by 17:00 IST on $d-1$; completed daily-history inputs are from $d-2$ or earlier. Weather summaries use archived forecasts with a documented 48-hour Previous Runs offset. Every admitted snapshot passes the availability guard. Historical observation publication times are unavailable, so local-history skill is conditional on the protocol's healthy-feed latency assumption.

B. Measured Targets and Sensor Quality

OpenAQ preserves the CPCB-provider station and sensor provenance [8]. Values must have convertible mass units, positive finite concentrations, at least 75% provider coverage and no provider rejection flag. Exact zeros and identical sequences of six or more consecutive bins are rejected under v3.1. Potential spikes and PM2.5 above PM10 remain flagged observations. Feature QC uses the prefix visible at the cutoff; later readings cannot revise an earlier feature. Rejected station-days remain in the dataset.

The common CPCB-style engine uses at least 16 distinct hours for each supported pollutant, at least three supported species and a required particulate species [9]. Particulates and designated gases use 24-hour means; CO and O_3 use the maximum complete trailing eight-hour mean, including seven preceding support hours. For a bounded breakpoint interval,

$$I_p(C) = I_L + \frac{I_H - I_L}{C_H - C_L}(C - C_L), \qquad I = \max_{p \in S} I_p(C_p). \quad (1)$$

The result is an AirGuard CPCB-style index, not an official CPCB bulletin. The Severe category is open-ended: $y \geq 401$ is a lower bound. Such targets are excluded from exact regression fitting and MAE/RMSE/bias; category, exceedance, below-401 and lower-bound shortfall diagnostics retain them.

C. Operational CAMS Reconstruction

The ADS dataset is the global atmospheric-composition forecast archive, not reanalysis [10]. GRIB type, producing date and time, lead hours, valid time, grid and units are decoded and verified. The downloaded grid is 0.4°; matching uses the frozen nearest grid centre. At the declared issue time the study uses the $d-1$ 00:00 UTC run, with nominal dissemination treated explicitly as an assumption rather than an empirical delivery log.

Surface PM mass concentrations are converted from kg m^{-3} to μg m^{-3}. Gas mass mixing ratio x uses the same run's lowest-model-level temperature T, specific humidity q and hybrid pressure p:

$$\rho = \frac{p}{287.05T(1 + 0.6078q)}, \qquad C_{\mu g/m^3} = 10^9 \rho x. \quad (2)$$

CO uses 10^6 for mg m^{-3}. Model-level fields are interpolated within their producing run before conversion. Missing brackets, hybrid coefficients or inconsistent metadata cause rejection. Retrieval time records acquisition provenance; it does not establish historical publication time.

D. Direct and Residual Calibration

Let $b_{s,d}$ denote the frozen raw CAMS AQI. Direct learners predict $y_{s,d}$; residual learners fit exact measured-minus-CAMS labels:

$$r_{s,d} = y_{s,d} - b_{s,d}, \qquad \widehat{y}_{s,d} = \max\{0, b_{s,d} + f_\theta(X_{s,d})\}. \quad (3)$$

Ridge uses standardised, training-imputed features with an L2 penalty. LightGBM and XGBoost provide nonlinear alternatives. Missing-value statistics and learned parameters are fitted only on training rows. Predictor groups comprise CAMS concentrations and descriptors, measured history, forecast weather, calendar information and spatial context. Fire features remain optional and require admissible publication evidence before evaluation. Target-specific pollutant support is an evaluation annotation, never a predictor. The current valid targets use the frozen PM2.5/PM10/O_3 support; the six-pollutant index is a secondary comparison representing the live product.

E. Uncertainty and Advisories

Split conformal calibration uses exact calibration residuals e_i with

$$k = \lceil (n+1)(1-\alpha) \rceil, \quad q_\alpha = e_{(k)}, \\ \mathcal{C}(x) = [\max(0, \widehat{y} - q_\alpha), \widehat{y} + q_\alpha]. \quad (4)$$

If the finite-sample rank cannot be supported, an interval is unavailable. Conformalised quantile LightGBM and forecast-category Mondrian calibration are additional declared methods. Membership and quantile-crossing repair use forecasts, never observed test categories. Censored-Severe compatibility ($U \geq 401$) is not identified coverage of the unknown true index.

Advisory policies use the point forecast (P0), calibrated median (P1), fixed 0.75 quantile (P2), or the smallest candidate quantile meeting a calibration-only under-warning target (P3). For observed category c and warning category w,

$$L_- = \mathbf{1}\{w < c\}, \quad L_+ = \mathbf{1}\{w > c\}, \quad \widehat{R}_- = \frac{k_- + 1}{n+1}. \quad (5)$$

TABLE I
FEATURE AVAILABILITY AND OPERATIONAL ASSUMPTIONS. STATIC METADATA AND THE CALENDAR DO NOT REQUIRE OUTCOME INFORMATION.

Group	Admitted information
A CAMS	Verified operational run, concentrations, sub-indices and diurnal descriptors
B History	Bins ending by 17:00 IST; completed daily lags from $d-2$; healthy-feed assumption
C Weather	Archived previous-run forecasts with pre-issue run bound
D Calendar	Target-day month, weekday and cyclical season information
E Spatial	Frozen coordinates, terrain and admissible pre-issue neighbours
F Fire	Optional; withheld without historical publication evidence

The declared targets are 10% for general adults and 5% for the other exposure strata. No feasible candidate is reported as such. Under dependence, these are empirical goals rather than a guaranteed personal or clinical risk bound.

F. Feature Availability Audit

The feature contract distinguishes a value's valid time from when it could have been available. A forecast can describe future valid hours while still being an admissible pre-issue input; an observation cannot. Retrieving an old field today is insufficient without its producing-run identity. The same distinction applies to archived weather, measured-history latency and fire publication. Table I documents the admitted groups and remaining assumptions.

G. Baseline Support and Missingness

The primary CAMS comparison does not credit a learner for removing a species the station failed to measure. Frozen PM2.5/PM10/O_3 support is defined from sensor metadata; the current valid join has no support mismatch. The six-pollutant index remains a secondary comparator because it is closer to the live product's display. Both are produced by the same engine, so target and baseline breakpoints, averaging windows and Severe treatment cannot drift.

History lags use station/date keys rather than row offsets. Missing days remain missing; neither targets nor forecast indices are borrowed from the wrong date. Censored historical values are not fabricated exact bias labels. Neighbour summaries use cutoff-compliant measurements and exclude self-donations. Monitor-free spatial regimes must remove unavailable donor feeds, not merely drop a station identifier. Missing predictors are imputed using training-only statistics with indicators, preserving the conditions under which a forecast could be issued.

H. Advisory Evaluation Contract

Under-warning and over-warning are evaluated jointly. Serious under-warning (two or more categories), misses on Poor-or-worse days and warning distance are separate diagnostics. P1 and P2 use one-sided calibration of their fixed quantiles; P3 selects among declared raw quantile forecasts on the calibration window. That selection is itself its calibration step. No test-window outcome can select a warning level or repair an interval.

The candidates are $\{0.5, 0.65, 0.75, 0.85, 0.9, 0.95, 0.975\}$. An absent feasible candidate is a failed empirical risk target, not an excuse to relabel a permissive warning as controlled. Shared station outcomes across the seven exposure strata are not seven independent clinical samples. The final report must retain unfavourable trade-offs, including cases where reduced under-warning requires substantial over-warning.

TABLE II
MEASURED DATASET AND CURRENT OPERATIONAL ALIGNMENT. COMPLETED MEASURED TARGETS ARE PRESERVED; ARCHIVE COVERAGE IS PARTIAL.

Measure	Count
Usable measured stations / cities	179 / 100
Retained measured station-days	107,280
Valid / rejected targets	64,799 / 42,481
Censored Severe targets	383
Guarded CAMS-aligned development rows	7,560
Aligned stations / cities / dates	180 / 100 / 42
Aligned valid / rejected targets	5,506 / 2,054
Verified raw archive chunks	6 / 36

IV. EXPERIMENTAL SETUP

A. Study Data and Provenance

The locked window is 19 February 2025–7 October 2026. The metadata-only sample selected 181 stations from an eligible catalogue of 481 stations in 255 parsed cities. Full-window counts in Table II are inherited descriptive metadata; final-test labels are not opened to produce this draft. All analysis discussed here is development-only.

B. Chronological and Spatial Splits

Expanding-window validation begins with 90 training dates, then 30 calibration dates and 30 evaluation dates, advancing by 30 dates. One-day embargoes separate adjacent partitions. Every station's row for a date moves together. Station eligibility requires 120 valid training targets, so the first 90-day training fold is necessarily insufficient; later outcomes cannot rescue it. The latest 90 dates, 10 July–7 October 2026, remain sealed.

Spatial validation trains through 31 October 2025 and tests from 2 November 2025 through the final pre-test day. Ten-fold 25-km urban-cluster and station holdouts, and six region holdouts, distinguish unseen cities from unseen stations in seen clusters. Local-history and monitor-free inference are reported separately. The all-station reference uses the same dates and test pairs. Unavailable held-out monitor feeds must be masked in the corresponding inference regime.

C. Baselines and Training-only Tuning

Climatology uses the station's training targets in the same month, with station and global means as fallbacks. Persistence uses the rolling measured 24-hour index ending at the cutoff.

TABLE III
DECLARED PRIMARY SEARCH SPACE. THESE ARE PREREGISTERED CANDIDATES, NOT SELECTED PARAMETERS.

Learner	Training-only choices
Ridge	$\alpha \in \{0.1, 1, 10, 100\}$
LightGBM	Leaves 15/31; minimum child samples 20/60
XGBoost	Depth 4/6; minimum child weight 5/20
Boosting controls	Rate 0.05; at most 600 rounds; early stop 40; row and column fractions 0.8

Raw CAMS uses the same AQI engine. The primary exact comparison requires all three baselines on identical pairs; secondary comparisons relax only persistence availability. Severe days remain in their category and lower-bound diagnostics.

Hyperparameters and early stopping use the last 30 dates of a fold's training window as inner validation. Calibration and evaluation labels do not tune the learner. All seeds are 42. The carry-forward model has the lowest mean rolling-fold MAE, with ties favouring Ridge and residual formulations; raw CAMS is retained if no learned model improves that criterion. Ablation fixes the learner while varying feature groups. Fire admission requires a favourable 95% paired date-block interval.

D. Metrics and Statistical Comparison

Exact-target MAE is primary; RMSE, signed forecast bias, coefficient of determination and six-category diagnostics are secondary. Poor-or-worse and Very-Poor-or-worse detection, false alarms and lower-bound Severe shortfall are reported explicitly. Regional, seasonal and city-size subgroups below 30 pairs retain counts but withhold inferential metrics. Intervals are assessed at 80% and 90% nominal coverage, with width and severity-specific coverage.

Paired comparisons preserve station/date pairs. Moving seven-day date blocks with 5,000 resamples provide percentile intervals; daily mean loss differences use a Diebold–Mariano comparison with Newey–West variance and Holm adjustment [11]. An interval spanning zero does not establish a nonzero difference. The immutable non-CAMS study uses a separate 30-valid-day station rule and fixed Ridge $\alpha = 10$; it cannot select the primary model.

V. RESULTS AND DISCUSSION

A. Preliminary Measured-data Evidence

The separate non-CAMS run contains 24 temporal evaluations (expanding and sliding sensitivities), 36 spatial holdouts and 12 calibration evaluations. Table IV and Fig. 2 show two paired development comparisons. Strict Ridge improves climatology on these pairs; history Ridge does not establish improvement over persistence. These comparisons answer neither the raw-CAMS correction question nor the locked final-test question.

B. Uncertainty and Extreme-event Failures

First-fold marginal interval coverage masks a high-pollution failure. Both nominal intervals cover none of the 35 exact Very Poor outcomes or 48 exact Poor-or-worse pairs. The individual Poor subgroup has only 13 pairs and its subgroup metric is withheld. Figure 3 retains these denominators rather than treating unavailable estimates as zeros.

TABLE IV
NON-CAMS PRELIMINARY PAIRED MAE DIFFERENCES. NEGATIVE DIFFERENCES FAVOUR RIDGE; INTERVALS USE SEVEN-DAY DATE BLOCKS.

Comparison	ΔMAE	95% interval	n
Strict Ridge–climatology	-8.96	[-12.49, -5.61]	38,382
History Ridge–persistence	+0.61	[-0.41, +1.70]	31,622

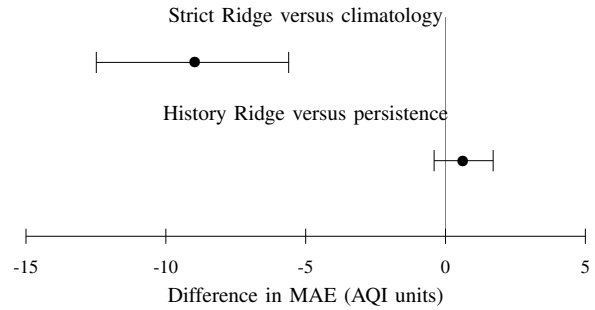


Fig. 2. Paired non-CAMS development differences. The persistence comparison crosses zero; a lower error than climatology does not establish a CAMS improvement.

C. Primary CAMS Status

The request manifest verifies 6 of 36 chunks. The guarded join covers 42 dates and 7,560 station-days, retaining 2,054 rejected targets and 2 censored Severe targets. This coverage cannot support the locked training and verification windows. No primary model has been fitted or selected. Missing archive months are not replaced with reanalysis, later operational fields or fabricated data. *Primary CAMS comparison: [RESULT PENDING CAMS ARCHIVE EXPERIMENT].*

D. Prospective Application

The audited ledger has 24 immutable forecasts across 2 target days and 0 verified days. 0 forecasts have pre-issue station bindings; 12 historical entries lack an issuance Git SHA. Missing historical identities or station bindings are not inferred after the outcome. The production application displays raw CAMS-based estimates; no experimental correction or unvalidated interval is deployed. Its interface and API checks establish software behaviour, while predictive skill requires matched outcomes and the declared research evaluation.

E. Calibration Stability Across Folds

Figure 4 displays mean interval widths for the existing non-CAMS folds. These are fold-level summaries, not a reconstructed histogram of individual intervals. Both width and coverage must be inspected: wider intervals can improve marginal coverage while retaining poor subgroup performance.

TABLE V
FIRST-FOLD NON-CAMS INTERVAL DIAGNOSTICS. WIDTH IS IN AQI UNITS; NEITHER ROW ESTABLISHES OPERATIONAL CALIBRATION.

Nominal	Exact coverage	Mean width	n
80%	0.855	95.5	2,923
90%	0.944	120.6	2,923

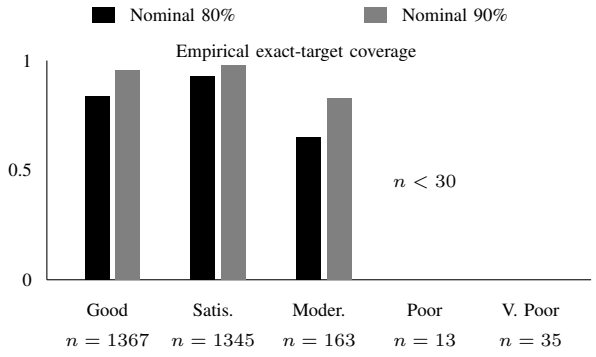


Fig. 3. Non-CAMS first-fold coverage by observed category. Both Very Poor estimates are zero; the Poor metric is unavailable at $n < 30$. Satisfactory and Moderate are abbreviated on the axis.

A censored Severe outcome identifies a lower bound, so whether an upper endpoint reaches 401 is reported separately from exact coverage.

F. Evidence Boundaries

The initial 90-day training fold cannot meet a station's 120-valid-day eligibility requirement. It is preserved as insufficient rather than rescued using later coverage. The preliminary non-CAMS study's separate 30-day station threshold and fixed penalty therefore must not be conflated with primary tuning. Likewise, paired regression evidence is restricted to exact targets; censored observations are retained for the declared category and lower-bound analysis.

The comparison with persistence is an important negative finding. A confidence interval containing zero does not demonstrate that two methods are identical, but it also does not support claiming a reliable difference. The first-fold Very Poor interval failure demonstrates why a single average coverage number cannot justify high-pollution warnings. These observations support the evaluation design, not a claim that station correction already improves genuine archived CAMS.

VI. LIMITATIONS

A. Availability and Representation

The current catalogue can introduce survivorship bias, and the measured sample does not establish nationwide representativeness. A coarse CAMS grid and a station describe different spatial supports. Historical station ingestion times and empirical per-run CAMS delivery logs are unavailable; nominal publication and healthy-feed assumptions remain explicit. Optional fire acquisition timestamps do not prove publication before issue.

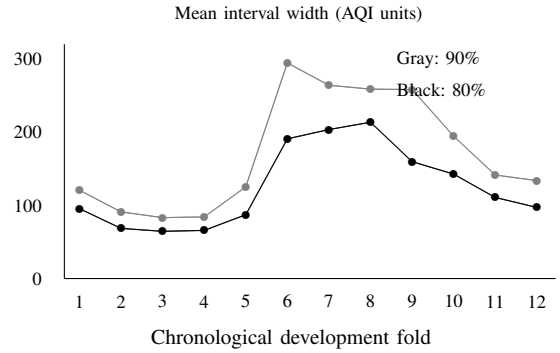


Fig. 4. Non-CAMS preliminary mean widths across chronological folds. The calibrated intervals remain development diagnostics rather than validated operational intervals.

B. Statistical and Advisory Scope

Preliminary measured-data gains cannot establish correction of raw CAMS. Temporal and spatial dependence limit textbook exchangeability arguments; coverage and warning-risk attainment are empirical findings. Rare high-pollution groups may be too small for stable inference. Warning groups specify exposure guidance tolerances, not diagnoses, prescriptions, personal disease probabilities or causal health benefits.

C. Outstanding Research Gates

Paired archive coverage, primary experiments, reproducibility and the actual model-selection lock remain incomplete. The final interval can be opened only after a committed configuration lock, a READY evidence audit and fresh human approval. Formatting and software deployment cannot release that gate.

VII. CONCLUSION AND FUTURE WORK

AirGuard provides a measured-station dataset, an audited operational forecast pipeline and a locked design for testing station correction and warning calibration. Existing non-CAMS evidence supports a gain over climatology but does not establish a gain over persistence, and shows severe subgroup failures despite useful marginal coverage. The main CAMS comparison and final-test results remain pending. Completing the archived cube will permit the declared temporal, spatial, uncertainty and advisory analyses; future prospective records must bind stations before issuance and accumulate real verified outcomes.

VIII. REPRODUCIBILITY AND DATA AVAILABILITY

The research branch provides acquisition manifests, QC definitions, feature guards, split utilities, experiment configurations and claim-evidence mappings. Raw and derived files have checksums; versioned caches are outside Git. The measured semantic hash begins `4761a5da60dde059`, the protocol hash `e24b664747029a76`, and the immutable preliminary run `287c87db04a6b294`. Full identifiers are supplied in the companion manifest. A measured-data reconstruction passed its recorded semantic audit; complete primary reproduction

remains pending. Current incomplete analyses are not represented by a fictitious model-selection lock.

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Claude Code and OpenAI Codex assisted software, tests and manuscript drafting. The author is responsible for verifying the methods, citations, results and disclosures before submission. No human-subject study or clinical validation is claimed. The supplied LogSight paper informs presentation structure only; its content, author list, model settings and numerical results are not reused.

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